

JACK HENDERSON

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EDUCATION

University of Edinburgh

Edinburgh, Scotland

09/2024 - 09/2025

- MSc (merit) - Particle and Nuclear Physics

Dissertation: The effect of Chern-Simons gravity on inflationary squeezed states in the early universe

Heriot-Watt University

Edinburgh, Scotland

09/2019 - 06/2023

- BSc (Hons) - Physics

Final Project: High-Dimensional entanglement verification via quantum Fisher information

RESEARCH EXPERIENCE

Institute of Photonics and Quantum Optics - paid research intern

Edinburgh, Scotland

August 2021

- * Investigated the theory of measurement-device-independent (MDI) entanglement verification for high-dimensional photonic systems
- * Produced an MDI entanglement witness (EW) and proved its validity for orbital angular momentum bipartite states
- * Implemented numerical analysis in MATLAB for dimension $D=2-10$ and documented results for the group

Quantum Optics and Computational Imaging group - paid research intern

Edinburgh, Scotland

July 2022

- * Developed a data-processing pipeline for a high-dimensional photonic experiment to perform full quantum state tomography
- * Integrated LabVIEW acquisition with MATLAB analysis, produced reconstructed states and summary plots for the group

Institute of Photonics and Quantum Optics - paid research intern

Edinburgh, Scotland

August 2022

- * Investigated an entanglement witness based on Quantum Fisher Information
- * Showed this EW outperforms standard targeted EWs for the states we were considering

Final Project

Heriot-Watt University

May 2023

- * Continued investigation of Quantum Fisher Information EW
- * modelled performance under different noise conditions, dimensionality and number of mutually unbiased measurements

Dissertation

The University of Edinburgh

October 2025

- * Studied the effect of a (constant- θ) Chern-Simons/Pontryagin term $\theta R\tilde{R}$ on inflationary tensor squeezing.
- * Derived helicity-dependent shifts to the squeezing for a constant θ , with potential implications for parity-odd CMB polarisation correlators
- * Ongoing: extending the project to decoherence of super-horizon tensor modes with $\theta R\tilde{R}$, manuscript in preparation with collaborators.

SKILLS

- C++, Python (with Scikit-learn/TensorFlow), MATLAB, LabVIEW